Lista 7 - MNED

1. A função u satisfaz a equação

$$x^2 u \frac{\partial u}{\partial x} + e^{-y}\frac{\partial u}{\partial y} = -u^2$$

e a condição $u = 1$ em $y = 0$, $0 < x < \infty$;

(a) Calcule a equação das características sobre o ponto $R(x_R, 0)$ com $x_R > 0$ e a solução sobre esta característica;

(b) Proponha um método de diferenças finitas para aproximar a solução desta equação. Discuta sobre sua escolha.

$$x^2 u u_x + e^{-y} u_y = -u^2$$

$$u = 1, \quad (x,y) \in (0,\infty)\times\{0\}$$

a) $\gamma_r(s) = (x_r(s), y_r(s)) \qquad \gamma_r(0) = (r, 0)$

$$u(\gamma_r(s))' = u_x(\gamma_r(s))\, x_r'(s) + u_y(\gamma_r(s))\, y_r'(s) = -u^2(\gamma_r(s))$$

$$\Rightarrow \begin{cases} x_r^2(s)\, u(\gamma_r(s)) = x_r'(s) \\ e^{-y_r(s)} = y_r'(s) \end{cases}$$

$$y_r'(s) = e^{-y_r(s)} \Rightarrow e^{y_r(s)} y_r'(s) = 1 \qquad y_r'(s)\, e^{y_r(s)} = \frac{d}{ds}\left(e^{y_r(s)}\right)$$

$$\Rightarrow \int \frac{d}{ds}\left(e^{y_r(s)}\right) ds = s + C \Rightarrow e^{y_r(s)} = s + C \Rightarrow y_r(s) = \ln(s + C)$$

$$y_r(0) = 0 \Rightarrow \ln(C) = 0 \Rightarrow C = 1$$

$$\therefore y_r(s) = \ln(s+1)$$

$$\frac{d}{ds}\left(u(\gamma_r(s))\right) = -u^2(\gamma_r(s)) \Rightarrow \begin{cases} w'(s) = -w^2(s) \\ w(0) = 1 \end{cases} \Rightarrow \frac{1}{w(s)} = s + K \Rightarrow$$

$$\Rightarrow u(\gamma_r(s)) = \frac{1}{s+1}$$

$$x_r'(s) = x_r^2(s)\, u(\gamma_r(s)) \Rightarrow \frac{x_r'(s)}{x_r^2(s)} = u(\gamma_r(s)) \Rightarrow \int \frac{x_r'(s)}{x_r^2(s)} ds = \int \frac{1}{s+1} ds$$

$$z = x_r(s), \quad dz = x_r'(s)ds \Rightarrow \int \frac{1}{z^2} dz = -\frac{1}{z} = \frac{-1}{x_r(s)} = \int \frac{1}{s+1} ds = \int \frac{1}{a} da = \ln(a) + M = \ln(s+1) + M$$

$$a = s+1, \quad da = ds$$

$$\Rightarrow x_r(s) = \frac{-1}{\ln(s+1) + M} \qquad x_r(0) = r \Rightarrow \frac{-1}{M} = r \Rightarrow M = \frac{-1}{r}$$

$$\Rightarrow x_r(s) = \frac{-1}{\ln(s+1) - \frac{1}{r}}$$

$$\therefore \gamma_r(s) = \left(\frac{-1}{\ln(s+1) - \frac{1}{r}}, \ln(s+1)\right) = (x_r(s), y_r(s))$$

Características

$$e^{y_r(s)} = s+1 \Rightarrow s = e^y - 1$$

$$\frac{-1}{x_r(s)} = \ln(s+1) - \frac{1}{r} \Rightarrow \frac{1}{r} = \ln(e^y - 1 + 1) + \frac{1}{x_r(s)} = y + \frac{1}{x_r(s)} = \frac{y x_r(s) + 1}{x_r(s)}$$

$$\Rightarrow r = \frac{x}{xy+1}$$

$$\boxed{u(x,y) = \frac{1}{e^y - 1 + 1} = e^{-y}}$$

b) Note que as características sempre andam para a direita. De fato,

$$x_r'(s) = x^2 u = x^2 e^{-y} \geq 0$$
$$y_r'(s) = e^{-y} \geq 0$$

Note que podemos interpretar $u(x,0) = 1$ como uma condição inicial para a EDP.

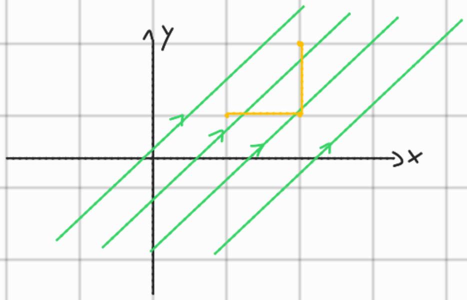
Podemos usar então upwind regressivo para x e progressivo para y quando crescemos y e regressivo para y quando o diminuímos.

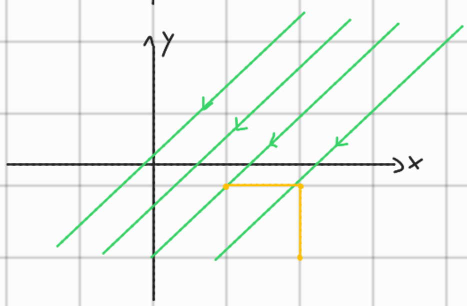
$$U_j^n \approx u(x_j, y_n)$$

$$x_j^2\, U_j^n \left(\frac{U_j^n - U_{j-1}^n}{\Delta x}\right) + e^{-y_n}\left(\frac{U_j^{n+1} - U_j^n}{\Delta y}\right) = -(U_j^n)^2$$

$$x_j^2\, U_j^n \left(\frac{U_j^n - U_{j-1}^n}{\Delta x}\right) + e^{-y_n}\left(\frac{U_j^n - U_j^{n-1}}{\Delta y}\right) = -(U_j^n)^2$$

Com isso obtemos um método explícito





2. Estude a convergência dos seguintes métodos para uma equação de advecção linear:

(a) Esquema explícito central;
(b) Lax-Friedrichs;
(c) Upwind;
(d) Leapfrog;
(e) Lax-Wendroff;
(f) Beam-Warming;

$u_t + a u_x = 0$

a) $U_j^{n+1} = U^n - \overbrace{\frac{a\Delta t}{2\Delta x}}^{\alpha} (U_{j+1}^n - U_{j-1}^n)$ $\quad O(\Delta t + \Delta x^2)$

Estabilidade:

$g(\omega) e^{i\omega j} = e^{i\omega j} - \alpha \left(e^{i\omega(j+1)} - e^{i\omega(j-1)} \right)$

$\Rightarrow g(\omega) = 1 - \frac{\alpha}{2i} \operatorname{sen}(\omega)$

$|g(\omega)|^2 = 1 + \alpha^2 \operatorname{sen}^2(\omega) \geq 1 \Rightarrow$ o método é sempre instável

$\therefore$ o método não é convergente

b) Lax-Friedrichs

$U_j^{n+1} = \frac{1}{2}\left(U_{j-1}^n + U_{j+1}^n\right) - \alpha\left(U_{j+1}^n - U_{j-1}^n\right)$

Reescrevendo:

$U_j^{n+1} = U_j^n + \frac{\Delta x^2}{2}\left(\frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{\Delta x^2}\right) - \Delta t \left(\frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x}\right)$

$\Rightarrow \frac{U_j^{n+1} - U_j^n}{\Delta t} = \frac{\Delta x^2}{2\Delta t}\left(\frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{\Delta x^2}\right) - \left(\frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x}\right)$

consistência: $\quad u(x_i, t_n) = u$

$\tau_j^n = u_t + O(\Delta t) - \frac{\Delta x^2}{2\Delta t}\left(u_{xx} + O(\Delta x^2)\right) - u_x + O(\Delta x^2)$

$= O(\Delta t + \Delta x^2) + \frac{\Delta x^2}{2\Delta t} u_{xx} = O\left(\Delta t + \frac{\Delta x^2}{\Delta t}\right)$

Se satisfaz a CFL, vemos que $\frac{\Delta x^2}{2\Delta t} = O(\Delta x)$

$\Rightarrow \tau_j^n = O(\Delta t + \Delta x)$

Estabilidade:

$g(\omega) e^{i\omega j} = \frac{1}{2}\left(e^{i\omega(j+1)} + e^{i\omega(j-1)}\right) - \alpha\left(e^{i\omega(j+1)} - e^{i\omega(j-1)}\right)$

$\Rightarrow g(\omega) = \cos(\omega) - 2i\alpha \operatorname{sen}(\omega)$

$|g(\omega)|^2 = \cos^2(\omega) + 4\alpha^2 \operatorname{sen}^2(\omega) \leq 1 \Rightarrow 4\alpha^2 \leq 1 = \frac{4a^2\Delta t^2}{4\Delta x^2} \leq 1 \Rightarrow \frac{|a|\Delta t}{\Delta x} \leq 1$

c) Upwind (apenas para $a > 0$)

$U_j^{n+1} = U_j^n - \overbrace{\frac{a\Delta t}{\Delta x}}^{\beta}\left(U_j^n - U_{j-1}^n\right)$

Consistência: $O(\Delta t + \Delta x)$

Estabilidade:

$g(\omega) = 1 - \beta(1 - e^{-i\omega})$

$= 1 - \beta + \beta\cos(\omega) - \beta i \operatorname{sen}(\omega) = 1 - \beta(1 - \cos(\omega)) - \beta i \operatorname{sen}(\omega) = 1 - 2\beta \operatorname{sen}^2\left(\frac{\omega}{2}\right) - \beta i \operatorname{sen}(\omega)$

$|g(\omega)|^2 = 1 - 4\beta \operatorname{sen}^2\left(\frac{\omega}{2}\right) + 4\beta^2 \operatorname{sen}^4\left(\frac{\omega}{2}\right) + \beta^2 \operatorname{sen}^2(\omega)$

$= 1 - 4\beta s + 4\beta^2 s^2 + 4s\beta^2(1-s) \leq 1 \qquad 0 \leq s \leq 1$

$\Rightarrow s^2 4\beta^2 (1-1) - 4\beta s + 4\beta^2 s \leq 0$

$\Rightarrow 4\beta^2 s \leq 4\beta s \Rightarrow \beta \leq 1 \Rightarrow \frac{a\Delta t}{\Delta x} \leq 1 \overset{a>0}{\Rightarrow} \frac{|a|\Delta t}{\Delta x} \leq 1$

d) Leapfrog

$U_j^{n+1} = U_j^{n-1} - \frac{a\Delta t}{\Delta x}\left(U_{j+1}^n - U_{j-1}^n\right)$

$\Rightarrow \frac{U_j^{n+1} - U_j^{n-1}}{2\Delta t} + a\,\frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x} = 0$

$\Rightarrow \tau_j^n = u_t + O(\Delta t^2) + a u_x + O(\Delta x^2) = O(\Delta t^2 + \Delta x^2)$

Estabilidade: $\xi = \xi^{-1} - \beta(e^{i\omega} - e^{-i\omega}) = \xi^{-1} - 2i\beta \operatorname{sen}(\omega) \Rightarrow$

$\xi^2 = 1 - 2i\beta \operatorname{sen}(\omega)\xi \Rightarrow \xi^2 + 2i\beta \operatorname{sen}(\omega)\xi - 1 = 0$

$\Delta = -4\beta^2 \operatorname{sen}^2(\omega) - 4(-1) = 4(1 - \beta^2 \operatorname{sen}^2\omega)$

$\xi_{1,2} = \frac{-\cancel{2}i\beta \operatorname{sen}(\omega) \pm \cancel{2}\sqrt{1 - \beta^2 \operatorname{sen}^2\omega}}{\cancel{2}}$ $\qquad \beta \operatorname{sen}\omega = \gamma$

$\xi_{1,2} = i\gamma \pm \sqrt{1 - \gamma^2}$

$|\gamma| = 1 \Rightarrow \xi_{1,2} = i\beta \operatorname{sen}(\omega) = |i\beta \operatorname{sen}(\omega)|^2 \leq 1 \Rightarrow \beta^2 \leq 1 \Rightarrow \frac{|a|\Delta t}{\Delta x} \leq 1$

$|\gamma| < 1 \Rightarrow |\xi_{1,2}|^2 = |i\gamma \pm \sqrt{1-\gamma^2}|^2 \leq 1 \Rightarrow \gamma^2 + 1 - \gamma^2 \leq 1 \Rightarrow 1 \leq 1$

$\Downarrow$

$\frac{|a|\Delta t}{\Delta x} < 1$

$|\gamma| > 1 \Rightarrow |\xi_{1,2}|^2 = |i\gamma \pm i\sqrt{\gamma^2 - 1}|^2 = |\gamma \pm \sqrt{\gamma^2 - 1}|^2 = \gamma^2 \pm 2\gamma\sqrt{\gamma^2 - 1} + \gamma^2 - 1 =$

$= 2\gamma^2 - 1 \pm 2\gamma\sqrt{\gamma^2 - 1} \leq 1 \Leftrightarrow \pm 2\gamma\sqrt{\gamma^2 - 1} \leq 2 - 2\gamma^2$

ξ_1: $\underbrace{2\gamma\sqrt{\gamma^2 - 1}}_{\geq 0} \leq 2 - 2\gamma^2 \Rightarrow 2 - 2\gamma^2 \geq 0 \Rightarrow \gamma^2 \leq 1 \Rightarrow \frac{|a|\Delta t}{\Delta x} \leq 1$ absurdo

$\therefore$ Região de estabilidade: $\frac{|a|\Delta t}{\Delta x} < 1$

e) Lax-Wendroff

$U_j^{n+1} = U_j^n - \frac{a\Delta t}{2\Delta x}\left(U_{j+1}^n - U_{j-1}^n\right) + \frac{a^2\Delta t^2}{2\Delta x^2}\left(U_{j+1}^n - 2U_j^n + U_{j-1}^n\right)$

$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\left(\frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x}\right) - \frac{a^2\Delta t}{2}\left(\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{\Delta x^2}\right) = 0$

$$\tau_j^n = u_t + \frac{\Delta t u_{tt}}{2} + O(\Delta t^2) + a u_x + O(\Delta x^2) - \frac{a^2 \Delta t}{2}\left(u_{xx} + \frac{\Delta x^2}{12} u_{xxxx} + O(\Delta x^4)\right)$$

$$u_{xx} = \frac{u_{tx}}{a} = \frac{u_{tt}}{a^2} \Rightarrow \tau_j^n = \frac{\Delta t}{2} u_{tt} - \frac{\Delta t}{2} u_{tt} + O(\Delta t^2 + \Delta x^2) = O(\Delta t^2 + \Delta x^2)$$

Estabilidade:

$$g(w) = 1 - \frac{a \Delta t}{2 \Delta x}(e^{iw} - e^{-iw}) + \frac{a^2 \Delta t^2}{2 \Delta x^2}(-2 + e^{iw} + e^{-iw}) =$$

$$= 1 - i\beta \operatorname{sen}(w) - \beta^2 (1 - \cos(w)) = 1 - 2\beta^2 \operatorname{sen}^2\left(\frac{w}{2}\right) - i\beta \operatorname{sen}(w)$$

$$|g(w)|^2 \leq 1 \Leftrightarrow 1 - 4\beta^2 s + 4\beta^4 s^2 + \beta^2 \cdot 4s(1-s) \leq 1 \qquad 0 \leq s = \operatorname{sen}^2\left(\frac{w}{2}\right) \leq 1$$

$$\Leftrightarrow 4\beta^2\left(-s + \beta^2 s^2 + s - s^2\right) \leq 0$$

$$\Leftrightarrow (\beta^2 - 1)s^2 \leq 0, \ \forall s \in [0,1]$$

$$\Leftrightarrow \beta^2 - 1 \leq 0 \Leftrightarrow \frac{|a|\Delta t}{\Delta x} \leq 1$$

f) Beam-warming $(a>0)$ $(a<0$ análogo$)$

$$U_j^{n+1} = U_j^n - \frac{a\Delta t}{2\Delta x}(3U_j^n - 4U_{j-1}^n + U_{j-2}^n) + \frac{a^2 \Delta t^2}{2\Delta x^2}(U_j^n - 2U_{j-1}^n + U_{j-2}^n)$$

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\left(\frac{3U_j^n - 4U_{j-1}^n + U_{j-2}^n}{2\Delta x}\right) - \frac{a^2 \Delta t}{2\Delta x^2}(U_j^n - 2U_{j-1}^n + U_{j-2}^n) = 0$$

$$\frac{3\cancel{u} - 4\left(\cancel{u} - \Delta x u_x + \frac{\Delta \cancel{x^2}}{2} u_{xx} + O(\Delta x^3)\right) + \cancel{u} - 2\Delta x u_x + 2\cancel{\Delta} x^2 u_{xx} + O(\Delta x^3)}{2\Delta x} =$$

$$= \frac{2\Delta x u_x + O(\Delta x^3)}{2\Delta x} = u_x + O(\Delta x^2)$$

$$\frac{\cancel{u} - 2\left(\cancel{u} - \Delta \cancel{x} u_x + \frac{\Delta x^2}{2} u_{xx} \overset{-\frac{\Delta x^3 u_{xxx}}{6}}{+} O(\Delta x^4)\right) + \cancel{u} - 2\Delta \cancel{x} u_x + 2\Delta x^2 u_{xx} \overset{-\frac{4}{3}\Delta x^3 u_{xxx}}{+} O(\Delta x^4)}{2\Delta x^2} =$$

$$= \frac{1}{2} u_{xx} - \frac{1}{2}\Delta x u_{xxx} + O(\Delta x^2)$$

$$\tau_j^n = \cancel{u_t} + \cancel{\frac{\Delta t}{2} u_{tt}} + \cancel{a u_x} + \frac{3a}{4}\Delta x^2 u_{xx} - \cancel{\frac{\Delta t}{2}} u_{tt} + \frac{\Delta x \Delta t}{2} u_{xxx} + O(\Delta t^2 + \Delta x^2)$$

se $\frac{|a|\Delta t}{\Delta x} \leq 1$, então $|a|\Delta t \leq \Delta x \Rightarrow \Delta x \Delta t = O(\Delta x^2)$

$$\Rightarrow \tau_j^n = O(\Delta t^2 + \Delta x^2)$$

Estabilidade $\qquad U_j^{n+1} = g(w) e^{ijw}$

$$g(w) = 1 - \frac{\beta}{2}(3 - 4e^{-iw} + e^{-2iw}) + \frac{\beta^2}{2}(1 - 2e^{-iw} + e^{-2iw})$$

$$\Rightarrow g(w)e^{iw} = e^{iw} - \frac{\beta}{2}(3e^{iw} - 4 + e^{-iw}) + \frac{\beta^2}{2}(\underbrace{e^{iw} - 2 + e^{-iw}}_{2\cos(w) - 2})$$

$$= e^{iw} - \frac{\beta}{2}(2\cos(w) + 2e^{iw} - 4) + \beta^2(\cos(w) - 1)$$

$$= e^{iw}(1-\beta) - \beta(\cos(w) - 2) + \beta^2(\cos(w) - 1) = e^{iw} + \beta\left(2\operatorname{sen}^2\left(\frac{w}{2}\right) + 1\right) - 2\beta^2 \operatorname{sen}^2\left(\frac{w}{2}\right)$$

$$e^{iw}(g(w) - 1) = \beta\left(2\operatorname{sen}^2\left(\frac{w}{2}\right) + 1\right) - 2\beta^2 \operatorname{sen}^2\left(\frac{w}{2}\right) =$$

$$= \beta + 2s\beta - 2\beta^2 s$$

$$\Rightarrow e^{iw} g(w) = e^{iw} + \beta + 2s\beta(1-\beta) = \cos(w) + i\operatorname{sen}(w) + \beta + 2s\beta(1-\beta)$$

$$|e^{iw} g(w)|^2 = |g(w)|^2 = (\cos(w) + \beta + 2s\beta(1-\beta))^2 + \operatorname{sen}^2(w)$$

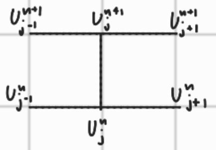
$$= \underbrace{\cos^2(w)}_{1 - \operatorname{sen}^2(w)} + 2\cos(w)\beta + \cos(w)4s\beta(1-\beta) + \beta^2 + 4s\beta^2(1-\beta) + 4s^2\beta^2(1-\beta)^2 + \cancel{\operatorname{sen}^2(w)}$$

$$= 1 + 2\cos(w)\beta + \cos(w)4s\beta(1-\beta) + \beta^2 + 4s\beta^2(1-\beta) + 4s^2\beta^2(1-\beta)^2$$

$$= 1 + 2(1-2s)\beta + (1-2s)4s\beta(1-\beta) + \beta^2 + 4s\beta^2(1-\beta) + 4s^2\beta^2(1-\beta)^2$$

3. Deduza um método para a equação de advecção, com a mesma molécula computacional do método de Crank-Nicolson. Quais são suas propriedades?

$$u_t + a u_x = 0$$



$$U_j'(t) = \frac{-a}{2\Delta x}\left(U_{j+1}(t) - U_{j-1}(t)\right)$$

Centramos em $n + \frac{1}{2}$

$$U_j^{n+1} = U_j^n - \frac{a\Delta t}{2\Delta x}\left[\frac{1}{2}\left(U_{j+1}^{n+1} - U_{j-1}^{n+1}\right) + \frac{1}{2}\left(U_{j+1}^n - U_{j-1}^n\right)\right]$$

Implícito

$$\frac{U^{n+1} - U_j^n}{\Delta t} + \frac{a}{2}\left(\frac{U_{j+1}^{n+1} - U_{j-1}^{n+1}}{2\Delta x}\right) + \frac{1}{2}\left(\frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x}\right) = 0$$

$$\tau_j^n = u_t + \frac{\Delta t}{2} u_{tt} + \frac{\Delta t^2}{6} u_{ttt} + O(\Delta t^3) + \frac{a}{2}\left(u_{x_j}^{n+1}\right) + \frac{a}{2}\left(u_{x_j}^n\right) + O(\Delta x^2)$$

$$u_{x_j}^{n+1} = u_x + \Delta t u_{xt} + \frac{\Delta t^2}{2} u_{xtt}$$

$$\tau_j^n = u_t + a u_x + \frac{a}{2}\left(\Delta t u_{xt} + \frac{\Delta t^2}{2} u_{xtt}\right) + \frac{\Delta t}{2} u_{tt} + O(\Delta x^2 + \Delta t^3) =$$

$$= -\frac{\Delta t}{2} u_{tt} + \frac{\Delta t}{2} u_{tt} + O(\Delta x^2 + \Delta t^2) = O(\Delta x^2 + \Delta t^2)$$

Estabilidade:

$$g(w) = 1 - \frac{a\Delta t}{4\Delta x}\left[g(w)\left(e^{iw} - e^{-iw}\right) + e^{iw} - e^{-iw}\right]$$

$$= 1 - \frac{a\Delta t}{4\Delta t}\left[2i g(w)\operatorname{sen}(w) + 2i\operatorname{sen}(w)\right]$$

$$= 1 - \frac{\beta}{4} \cdot 2i\operatorname{sen}(w)\left(g(w) + 1\right)$$

$$\Rightarrow g(w) = \frac{1 - \frac{\beta}{2} i\operatorname{sen}(w)}{1 + \frac{\beta}{2} i\operatorname{sen}(w)}$$

$$\left|1 - \frac{\beta}{2} i\operatorname{sen}(w)\right|^2 = 1 + \frac{\beta^2}{4}\operatorname{sen}^2(w) =$$

$$= \left|1 + \frac{\beta}{2} i\operatorname{sen}(w)\right|^2$$

$$\Rightarrow |g(w)| = 1$$

$\therefore$ o método é incondicionalmente estável.

4. O *esquema implícito de Wendroff*, também conhecido com método *Box* para uma equação de advecção linear

$$u_t + au_x = 0, \quad u(x,0) = f(x), \quad u(0,t) = g(t), \quad f(0) = g(0)$$

com $a > 0$, é dado por

$$U_{j+1}^{n+1} = U_j^n + \frac{1-\nu a}{1+\nu a}(U_{j+1}^n - U_j^{n+1})$$

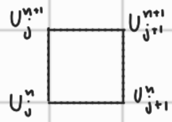
onde $\nu = k/h$.

(a) Desenhe a molécula computacional do método;

(b) Analise sua convergência e discuta sobre a condição CFL;

(c) Explique como este método implícito pode ser aplicado sem a resolução de sistemas lineares a cada passo de tempo.

a)



b) $(U^{n+1}_{j+1} - U^n_j)(1+\nu a) = (U^n_{j+1} - U^{n+1}_j)(1-\nu a)$

$$U^{n+1}_{j+1} - U^n_j = U^n_{j+1} - U^{n+1}_j - \nu a\left(U^{n+1}_{j+1} - U^n_j + U^n_{j+1} - U^{n+1}_j\right)$$

$$U^{n+1}_{j+1} - U^n_{j+1} + U^{n+1}_j - U^n_j + \nu a\left(U^{n+1}_{j+1} - U^{n+1}_j + U^n_{j+1} - U^n_j\right) = 0$$

$$\frac{U^{n+1}_{j+1} - U^n_{j+1}}{\Delta t} + \frac{U^{n+1}_j - U^n_j}{\Delta t} + \frac{a}{\Delta x}\left(U^{n+1}_{j+1} - U^{n+1}_j\right) + \frac{a}{\Delta x}\left(U^n_{j+1} - U^n_j\right) = 0$$

$$\tau^n_j = (u_t)^n_{j+1} + \frac{\Delta t}{2}(u_{tt})^n_{j+1} + (u_t)^n_j + \frac{\Delta t}{2}(u_{tt})^n_j + \frac{a}{\Delta x}\left((u_x)^{n+1}_j + \frac{\Delta x}{2}(u_{xx})^{n+1}_j\right) +$$

$$\frac{a}{\Delta x}\left((u_x)^n_j + \frac{\Delta x}{2}(u_{xx})^n_j\right) + O(\Delta t^2 + \Delta x^2) = \cdots = O(\Delta t^2 + \Delta x^2)$$

Estabilidade:

$$g(w)e^{iw} = 1 + \frac{1-\nu a}{1+\nu a}\left(e^{iw} - g(w)\right)$$

$$g(w) = \frac{1 + \frac{1-\nu a}{1+\nu a}e^{iw}}{e^{iw} + \frac{1-\nu a}{1+\nu a}} = \frac{1+\theta e^{iw}}{e^{iw}+\theta}$$

$$|1+\theta e^{iw}|^2 = 1 + 2\theta\cos(w) + \theta^2\cos^2(w) + \theta^2\text{sen}^2(w) = 1 + \theta^2 + 2\theta\cos(w)$$

$$|e^{iw}+\theta|^2 = \theta^2 + 2\theta\cos(w) + \cos^2(w) + \text{sen}^2(w) = 1 + \theta^2 + 2\theta\cos(w)$$

$\Rightarrow |g(w)|^2 = 1$

∴ o método é incondicionalmente estável

⇒ sempre convergente.

Note que o domínio de dependência numérico são todos os pontos (a esquerda) em um determinado tempo, o que implica que todos os pontos já calculados estão no domínio de dependência numérico. Portanto, não precisa satisfazer a condição de CFL.

c) $U^{n+1}_{j+1} = U^n_j + \frac{1-\nu a}{1+\nu a}\left(U^n_{j+1} - U^{n+1}_j\right)$

Note que $u(0,t) = g(t)$. Assim, $U^{n+1}_0 = g(t_{n+1})$. Com isso, podemos calcular explicitamente U^{n+1}_1. Podemos fazer isso indutivamente até U^{n+1}_m

5. Considere o problema

$$u_t + au_x = 0, \quad u(x,0) = f(x), \quad u(0,t) = g(t), \quad f(0) = g(0)$$

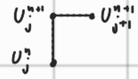
com a constante. Analise a estabilidade dos métodos implícitos

(a) $U_j^n = (1-\nu a)U_j^{n+1} + \nu a U_{j+1}^{n+1}$

(b) $U_j^{n+1} + \frac{\nu a}{4}(U_{j+1}^{n+1} - U_{j-1}^{n+1}) = U_j^n - \frac{\nu a}{4}(U_{j+1}^n - U_{j-1}^n)$

sendo $\nu = k/h$. O que dizer sobre a condição CFL para métodos implícitos?

a) Von-Neumann



$$1 = (1-\nu a)g(w) + \nu a g(w)e^{iw} \Rightarrow$$

$$1 = g(w)\left(1 - \nu a + \nu a e^{iw}\right) \Rightarrow g(w) = \frac{1}{1 - \nu a + \nu a e^{iw}}$$

$$|1 + \nu a(e^{iw} - 1)| = |1 - \nu a(1 - \cos(w)) + \nu a i\,\text{sen}(w)| =$$

$$= 1 - 2\nu a(1 - \cos(w)) + \nu^2 a^2(1 - 2\cos(w) + \cos^2(w)) + \nu^2 a^2 \text{sen}^2(w) =$$

$$1 + 2\nu^2 a^2 - 2\nu a\left(1 - \cos w + \nu a\cos(w)\right)$$

$$\Rightarrow \nu a \geq \left(2\text{sen}^2\left(\tfrac{w}{2}\right) + \nu a\cos(w)\right) = (2 - \nu a)\text{sen}^2\left(\tfrac{w}{2}\right) + \cos^2\left(\tfrac{w}{2}\right)$$

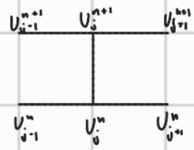
$$\cos(w) = \cos^2\left(\tfrac{w}{2}\right) - \text{sen}^2\left(\tfrac{w}{2}\right)$$

$$\Rightarrow \nu a \geq 2 - \nu a \Rightarrow \nu a \geq 1$$

Se $\nu a \geq 1$, o método é estável

b) $g(w) + \frac{\nu a}{4}\left(g(w)e^{iw} - g(w)e^{-iw}\right) = 1 - \frac{\nu a}{4}\left(e^{iw} - e^{-iw}\right)$

$$\Rightarrow g(w)\left(1 + \frac{\nu a}{2} i\,\text{sen}(w)\right) = 1 - \frac{\nu a}{2} i\,\text{sen}(w)$$



$$g(w) = \frac{1 - \frac{\nu a}{2} i\,\text{sen}(w)}{1 + \frac{\nu a}{2} i\,\text{sen}(w)} \Rightarrow |g(w)| = 1$$

∴ sempre estável.

A condição CFL está relacionada com o domínio de dependência. No item a), a condição CFL é:

$$\frac{|a|\Delta t}{\Delta x} \geq 1$$

O oposto dos métodos explícitos.

Para o item b), o método tem como domínio de dependência numérico todos os pontos anteriores, logo não precisa de CFL.

6. Considere a equação

$$u_t + a(x,t)u_x = 0, \quad x \geq 0, \quad t \geq 0$$

onde

$$a(x,t) = \frac{1+x^2}{1+2xt+2x^2+x^4}.$$

com condição inicial $u(x,0) = \psi(x)$.

(a) Mostre que a solução exata desta equação hiperbólica é

$$u(x,t) = \psi\left(x - \frac{t}{1+x^2}\right)$$

(b) Escreva um programa em Matlab para resolver esta equação usando os métodos Upwind, Lax-Wendroff, Box, Leapfrog e esquema implícito de primeira ordem (diferenças regressivas no tempo e no espaço);

(c) Resolva a equação acima utilizando as seguintes condições iniciais:

i.

$$\psi(x) = \begin{cases} 1, & 0.2 \leq x \leq 0.4 \\ 0, & \text{caso contrário.} \end{cases}$$

ii.

$$\psi(x) = e^{-10(4x-1)^2}$$

(d) Compare as soluções obtidas com malhas $h = 0.01$ e $h = 0.02$, nos tempos $t = 0$, $t = 0.1$, $t = 0.5$ e $t = 1$;

(e) Conduza testes de convergência numérica para os métodos implementados.

a) $\dfrac{\partial u}{\partial t}(x,t) = \Psi'\left(x - \dfrac{t}{1+x^2}\right)\cdot\dfrac{-1}{1+x^2}$

$\dfrac{\partial u}{\partial x}(x,t) = \Psi'\left(x - \dfrac{t}{1+x^2}\right)\left(1 + \dfrac{2xt}{(1+x^2)^2}\right)$

$a(x,t)\dfrac{\partial u}{\partial x}(x,t) = \left(\dfrac{(1+x^2)^2+2xt}{(1+x^2)^2}\right)\Psi'\cdot\left(\dfrac{1+x^2}{(1+x^2)^2+2xt}\right) =$

$= \dfrac{\Psi'}{1+x^2} = -\dfrac{\partial u}{\partial t}(x,t$

$u(x,0) = \Psi\left(x - \dfrac{0}{1+x^2}\right) = \Psi(x)$

∴ é uma solução

7. Seja a equação

$$u_{xx} = u^2 u_{tt}, \quad u(x,0) = 1 + x^2, \quad u_t(x,0) = 0.$$

(a) Construa um método usando diferenças centrais;

(b) É possível analisar a estabilidade deste método pelo critério de von Neumann? Se sim, desenvolva-o. Se não, como você faria?

a) $\dfrac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{\Delta x^2} = (U_j^n)^2\left(\dfrac{U_j^{n-1} - 2U_j^n + U_j^{n+1}}{\Delta t^2}\right)$

$U_j^{n+1} = 2U_j^n - U_j^{n-1} + \dfrac{\Delta t^2}{\Delta x^2 (U_j^n)^2}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right)$

Para o primeiro passo temporal

$U_j^1 = 2U_j^0 - U_j^{-1} + \dfrac{\Delta t^2}{\Delta x^2 (U_j^0)^2}\left(U_{j-1}^0 - 2U_j^0 + U_{j+1}^0\right)$ e $0 = \dfrac{\partial u}{\partial t}(x_j,0) \approx \dfrac{U_j^1 - U_j^{-1}}{2\Delta t}$ $O(\Delta t^2)$

$\Rightarrow U_j^{-1} = U_j^1$

$\therefore U_j^1 = U_j^0 + \dfrac{\Delta t^2}{2\Delta x^2 (U_j^0)^2}\left(U_{j-1}^0 - 2U_j^0 + U_{j+1}^0\right)$

b) Não podemos usar Von-Neumann, pois o método não é linear.

Nesse caso, linearizamos a equação

$v(x,t) + u^* = u(x,t)$

$\Rightarrow v(x,t) = u(x,t) - u^*$

$\dfrac{\partial^2 v}{\partial t^2} = \dfrac{\partial^2 u}{\partial t^2}$ e $\dfrac{\partial^2 v}{\partial x^2} = \dfrac{\partial^2 u}{\partial x^2} \Rightarrow v_{xx} = (v+u^*)^2 v_{tt}$

Se v fica próximo de u^*, então $v_{xx} \approx u^{*2} v_{tt}$

então aplicamos Von-Neumann na linearizada

$f(u^*+v) = (u^*+v)^2 = f(u^*) + v f'(u^*) + \dfrac{v^2}{2} f''(u^*) + \dfrac{v^3}{6} f'''(u^*) + \cdots =$

$= (u^*)^2 + v\,2u^* + v^2 + 0 = (u^*)^2 + O(v)$

8. Derive um esquema explícito, usando diferenças centrais para resolver

$$u_{xx} - (1+4x)^2 u_{tt} = 0, \quad 0 < x < 1, \quad t > 0$$

com

$$u(x,0) = x^2, \quad u_t(x,0) = 0, \quad u_x(0,t) = 0, \quad u(1,t) = 1.$$

(a) Estabeleça as condições CFL para esse método.

$\dfrac{1}{\Delta x^2}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right) - \dfrac{(1+4x_j)^2}{\Delta t^2}\left(U_j^{n-1} - 2U_j^n + U_j^{n+1}\right) = 0$ $\quad O(\Delta t^2 + \Delta x^2)$

$U_j^{n+1} = 2U_j^n - U_j^{n-1} + \dfrac{\Delta t^2}{\Delta x^2}\dfrac{1}{(1+4x_j)^2}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right)$

$u_t(x_j,0) = 0 \Rightarrow \dfrac{U_j^1 - U_j^{-1}}{2\Delta t} = 0 \Rightarrow U_j^{-1} = U_j^1$

Para $t = 0$

$U_j^1 = U_j^0 + \dfrac{\Delta t^2}{2\Delta x^2}\dfrac{1}{(1+4x_j)^2}\left(U_{j-1}^0 - 2U_j^0 + U_{j+1}^0\right) = (x_j)^2 + \dfrac{\Delta t^2}{2\Delta x^2}\dfrac{1}{(1+4x_j)^2}\left((x_{j-1})^2 - 2(x_j)^2 + (x_{j+1})^2\right)$

Nas bordas: $x = 0$

$\dfrac{U_1^n - U_{-1}^n}{2\Delta x} = 0 \Rightarrow U_1^n = U_{-1}^n = U_0^{n+1} = 2U_0^n - U_0^{n-1} + \dfrac{\Delta t^2}{\Delta x^2}\left(2U_1^n - 2U_0^n\right)$

$x = 1$

$U_m^n = 1$

a) Lembremos que se $u_{tt} = c^2 u_{xx}$, então podemos reescrever o problema como 2 advecções com $|a| = c$. Logo, devemos ter

$\dfrac{c\Delta t}{\Delta x} \leq 1.$

Nesse caso, $c = \dfrac{1}{1+4x} \Rightarrow \dfrac{\Delta t}{\Delta x} \leq 1 + 4x, \quad \forall x \in [0,1]$

$\boxed{\therefore \dfrac{\Delta t}{\Delta x} \leq 1 + 0x = 1}$

9. A equação

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$$

é aproximada no ponto $(j\Delta x, n\Delta t)$ pelo esquema implícito

$$\frac{1}{\Delta t^2}\delta_t^2 U_j^n = \frac{1}{\Delta x^2}\left(\frac{1}{4}\delta_x^2 U_j^{n+1} + \frac{1}{2}\delta_x^2 U_j^n + \frac{1}{4}\delta_x^2 U_j^{n-1}\right)$$

(a) Use o método de Von Neumann para estudar sua estabilidade;

(b) Encontre o erro de truncamento desse método.

a)

$$g(\omega) - 2 + \frac{1}{g(\omega)} = \frac{\Delta t^2}{\Delta x^2}\left[\frac{g(\omega)}{4}\left(e^{i\omega} + e^{-i\omega} - 2\right) + \frac{1}{2}\left(e^{i\omega} - e^{-i\omega} - 2\right) + \frac{1}{4g(\omega)}\left(e^{i\omega} - e^{-i\omega} - 2\right)\right] \Rightarrow$$

$$g^2(\omega) - 2g(\omega) + 1 = \frac{\Delta t^2}{\Delta x^2}\, 2(\cos(\omega) - 1)\left(\frac{g^2(\omega)}{4} + \frac{g(\omega)}{2} + \frac{1}{4}\right) \; \ldots\ldots$$

... incondicionalmente estável. Confie!

b) Nem fodendo!

10. Considere a equação da onda

$$u_{tt} = u_{xx}, \quad 0 < x < 1, \quad t > 0$$

com condições iniciais e de fronteira dadas por

$$u(x,0) = 100x^2, \quad u_t(x,0) = 200x, \quad u_x(0,t) = 200t, \quad u(1,t) = 100(1+t)^2$$

cuja solução exata é $u(x,t) = 100(x+t)^2$.

(a) Mostre que a aplicação do método de diferenças centrais, com a escolha $k = h = 1/2$, resulta na solução numérica $U_j^n = 100(x_j + t_n)^2$;

(b) Transforme a equação acima em um sistema de duas equações de transporte linear;

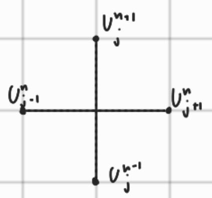
(c) Usando a decomposição espectral do sistema resultante, descreva como aplicar os métodos Upwind, Lax-Friedrichs, Lax-Wendroff, Leapfrog e Box. Não esqueça de descrever as mudanças adequadas nas condições iniciais e de contorno;

(d) Implemente estes métodos em Matlab e conduza testes numéricos, incluindo análise de convergência.

a) $U_j^{n+1} = 2U_j^n - U_j^{n-1} + \frac{\Delta t^2}{\Delta x^2}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right)$

$$\cancel{4}\left(U_j^{n-1} - 2\cancel{U_j^n} + U_j^{n+1}\right) = \cancel{4}\left(U_{j-1}^n - 2\cancel{U_j^n} + U_{j+1}^n\right)$$

$$U_j^{n+1} = -U_j^{n-1} + U_{j-1}^n + U_{j+1}^n$$



Para t_1:

$$\frac{U_j^1 - U_j^{-1}}{2\Delta t} = 200x_j \Rightarrow U_j^{-1} = U_j^1 - 200x_j$$

$$U_j^1 = 200x_j - U_j^1 + U_{j-1}^0 + U_{j+1}^0 \Rightarrow U_j^1 = 100x_j + 50\left((x_{j-1})^2 + (x_{j+1})^2\right) =$$

$$= 100x_j + 50\left((x_j - \tfrac{1}{2})^2 + (x_j + \tfrac{1}{2})^2\right) = 100x_j + 50\left(x_j^2 - x_j + \tfrac{1}{4} + x_j^2 + x_j + \tfrac{1}{4}\right)$$

$$= 100x_j + 100x_j^2 + 25 = 100\left(x_j + x_j^2 + \tfrac{1}{4}\right) = 100\left(x_j + \tfrac{1}{2}\right)^2 = 100\left(x_j + t_1\right)^2$$

Para x_0:

$$\frac{U_1^n - U_{-1}^n}{2\Delta x} = 200t_n \Rightarrow U_{-1}^n = U_1^n - 200t_n$$

$$U_0^{n+1} = -U_0^{n-1} + 2U_1^n - 200t_n$$

x_0, t_1:

$$U_0^1 = -U_0^1 + 200\cancel{t_0}^{\,0} + 2\cdot U_1^0 - 200\cancel{t_0}^{\,0}$$

$$\Rightarrow U_0^1 = U_1^0 = 200x_1^2 = 100\cdot(x_1 + t_0)^2$$

Caso geral

$$U_j^{n+1} = 100\left(x_j + t_n + \tfrac{1}{2}\right)^2$$

$$U_j^{n-1} = 100\left(x_j + t_n - \tfrac{1}{2}\right)^2$$

$$U_{j+1}^n = 100\left(x_j + t_n + \tfrac{1}{2}\right)^2$$

$$U_{j-1}^n = 100\left(x_j + t_n - \tfrac{1}{2}\right)^2$$

$$U_j^{n-1} = U_{j-1}^n = 100\left(x_j + t_n - \tfrac{1}{2}\right)^2$$

$$\Rightarrow -U_j^{n-1} + U_{j-1}^n = 0$$

$$\text{E } U_j^{n+1} = U_{j+1}^n$$

$$\Rightarrow U_j^{n+1} = U_{j+1}^n + U_{j-1}^n - U_j^{n-1}$$

b) $u_{tt} = u_{xx}$

$V = U_t \quad W = U_x \Rightarrow \quad V_x = U_{xt} \quad W_t = U_{tx}$

$V_t = U_{tt} \quad W_x = U_{xx}$

$$\begin{cases} V_t = W_x \\ V_x = W_t \end{cases} \Rightarrow \begin{cases} V_t - W_x = 0 \\ V_x - W_t = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} V_t \\ W_t \end{bmatrix}}_{V_t} + \underbrace{\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} V_x \\ W_x \end{bmatrix}}_{V_x} = 0$$

diagonalizando A: $\quad A = R\Lambda R^{-1}$

$$\det\begin{pmatrix} -\lambda & -1 \\ -1 & -\lambda \end{pmatrix} = 0 \Rightarrow \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$(A - Id)x = 0 \Rightarrow \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \Rightarrow \begin{cases} -x_1 - x_2 = 0 \\ -x_1 - x_2 = 0 \end{cases} \Rightarrow x_1 = -x_2$$

$$(A + Id)x = 0 \Rightarrow \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \Rightarrow x_1 = x_2$$

$b + d = 1$

$-a + c = 1 \Rightarrow 2c = 1$

$a = -c$

$b = d$

$$\therefore R = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow RR^{-1} = \begin{pmatrix} -a + c & -b + d \\ a + c & b + d \end{pmatrix} \Rightarrow R^{-1} = \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$\therefore \quad V_t + AV_x = 0 \Rightarrow R^{-1}V_t + R^{-1}R\Lambda R^{-1}V_x = 0 \Rightarrow R^{-1}V_t + \Lambda R^{-1}V_x = 0$$

$$\Rightarrow (R^{-1}V)_t + \Lambda(R^{-1}V)_x = 0$$

$$Z = R^{-1}V \Rightarrow Z_t + \Lambda Z_x = 0$$

$$\begin{cases} (z_1)_t + (z_1)_x = 0 \\ (z_2)_t - (z_2)_x = 0 \end{cases}$$

CI:

$W = u_x \Rightarrow W = 200x \Rightarrow w(x,0) = 200x$

$V = u_t \Rightarrow v(x,0) = 200x$

$$V_0 = \begin{pmatrix} 200x \\ 200x \end{pmatrix}$$

$$Z = R^{-1}V \Rightarrow Z_0 = R^{-1}V_0 = \frac{1}{2}\begin{pmatrix} 0 \\ 400x \end{pmatrix} = \begin{pmatrix} 0 \\ 200x \end{pmatrix}$$

$$Z(x,0) = \begin{pmatrix} 0 \\ 200x \end{pmatrix}$$

Cond. contorno

$u_x(0,t) = 200t$ $\qquad V = u_t \Rightarrow v(1,t) = 200(1+t)$

$u(1,t) = 100(1+t)^2$

$\qquad w = u_x \Rightarrow w(0,t) = 200t$

$u_{tx}(0,t) = 200$

$$Z(0,t) = R^{-1}V(0,t) = \frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} v(0,t) \\ w(0,t) \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} v(0,t) \\ 200t \end{pmatrix}$$

$$= \begin{pmatrix} \frac{-v(0,t)}{2} + 100t \\ \frac{v(0,t)}{2} + 100t \end{pmatrix} = z_1(0,t) + z_2(0,t) = 200t$$

$$Z(1,t) = R^{-1}V(1,t) = \frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 200(1+t) \\ w(1,t) \end{pmatrix} = \begin{pmatrix} -100(1+t) + w(1,t)/2 \\ 100(1+t) + w(1,t)/2 \end{pmatrix}$$

$\Rightarrow z_2(1,t) - z_1(1,t) = 200(1+t)$

$$\therefore \text{C.C.:} \begin{cases} z_1(0,t) + z_2(0,t) = 200t \\ z_2(1,t) - z_1(1,t) = 200(1+t) \end{cases}$$

c) Upwind:

$$\Lambda^+ = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \qquad \Lambda^- = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \qquad \Lambda = \Lambda^+ + \Lambda^-$$

$$\Rightarrow Z_t(x,t) + (\Lambda^+ + \Lambda^-) Z_x(x,t) = 0$$

$$\therefore \frac{Z_j^{n+1} - Z_j^n}{\Delta t} = -\frac{\Lambda^+}{\Delta x}(Z_j^n - Z_{j-1}^n) - \frac{\Lambda^-}{\Delta x}(Z_{j+1}^n - Z_j^n)$$

com $Z_j^n = \begin{bmatrix} (z_1)_j^n \\ (z_2)_j^n \end{bmatrix}$

$$Z_j^{n+1} = Z_j^n - \frac{\Delta t}{\Delta x}\left(\Lambda^+(Z_j^n - Z_{j-1}^n) + \Lambda^-(Z_{j+1}^n - Z_j^n)\right)$$

Na fronteira, conseguimos atualizar um dos valores. Colocamos o outro em função deste.

$$\text{sen}^2(\omega) = 1 - \cos^2(\omega) = 1 - \left(\cos^2\left(\frac{\omega}{2}\right) - \text{sen}^2\left(\frac{\omega}{2}\right)\right)^2 = 1 - \left(1 - \text{sen}^2\left(\frac{\omega}{2}\right) - \text{sen}^2\left(\frac{\omega}{2}\right)\right)^2$$

$$= 1 - (1-2s)^2 = 1 - (1 - 4s + 4s^2)$$

$$= 4s(1-s)$$

$$\cos(\omega) = \cos^2\left(\frac{\omega}{2}\right) - \text{sen}^2\left(\frac{\omega}{2}\right)$$

$$1 = \cos^2\left(\frac{\omega}{2}\right) + \text{sen}^2\left(\frac{\omega}{2}\right)$$

$$1 - \cos(\omega) = 1 - \left(\cos^2\left(\frac{\omega}{2}\right) - \text{sen}^2\left(\frac{\omega}{2}\right)\right) = 2\,\text{sen}^2\left(\frac{\omega}{2}\right)$$

$$\cos(\omega) = \cos^2\left(\frac{\omega}{2}\right) - \text{sen}^2\left(\frac{\omega}{2}\right) = 1 - 2\,\text{sen}^2\left(\frac{\omega}{2}\right) = 1 - 2s$$